Sets of Large measure.

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October 26, 2026

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III- Playing with k-thresholds: some simple and cute results.

I thank KIAS for having invited Jinyoung Park at the same time that I visited last year, which made these results possible.

Basic idea of concentration of measure

For many probability measures P on a space of LARGE DIMENSION the following occurs: Given any set A with $\mathrm{P}(A) \geq 1/2$ them most of the points of the space are close to A.

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"A point x is close to A if it is close to at least a point of A." Statement of the concentration of measure phenomenon: For any set $A \subset X$ with $\mathrm{P}(A) \geq 1/2$ then $\mathrm{P}(x; d(x, A) \geq t)$ is small for large t.

Consequences in terms of Lipschitz functions.

A function f on a metric space (X, d) is 1-Lipschitz if

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Under concentration of measure, 1-Lipschitz functions are nearly constant, equal to their median. Let M be a median of f:

$$\mathrm{P}(f \geq M) \geq 1/2; \mathrm{P}(f \leq M) \geq 1/2.$$

When there is concentration of measure, a1-Lipschitz function is "nearly constant equal to its median" because

$$f(x) \geq M + t \Rightarrow d(x, \{f \leq M\}) \geq t\}$$

and for large t the last set has small measure because $\mathrm{P}(\{f \leq M\}) \geq 1/2$.

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If $\mathrm{P}(A) \geq 1/2$ the median M of f is 0 and thus $\mathrm{P}(d(x, A) \geq t) = \mathrm{P}(f \geq M + t)$.

Fundamental example: the sphere S_N

$$S_N(1) = \{x \in \mathbb{R}^N; \|x\| = d(x,0) = 1\}.$$

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1-Lipschitz functions fluctuate by at most order $1/\sqrt{N}$. The isoperimetric inequality is much more precise than concentration of measure. The case of $S_N(1)$ generalizes to many other manifolds under conditions on the curvature.

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γ_N is invariant by rotations. It is "round".

γ_N is a product measure because

$$c^N \exp(-\|x\|^2) = \prod_{j\le N} c\exp(-x_j^2/2)$$

so that $\gamma_N = \gamma_1^{\otimes N}$. In some sense, γ_N is a "cube". The coordinates functions $x = (x_j)_{j\le N} \mapsto x_i$ are independent of law γ_1.

The Gaussian isoperimetric inequality

For $a > 0$, $t > 0$ and a set $A \subset \mathbb{R}^N$ with $\gamma_n(A) = a$, the maximum value of $\gamma_N(d(x, A) \geq t)$ is obtained when A is a half-space.

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Not caring about constants let us try to remember the following which we will use later:

$$\gamma_N(A) \geq 1/2 \Rightarrow \gamma_N(d(x, A) \geq t) \leq 2\exp(-t^2/2).$$

Relationship between the two examples.

Providing $\mathbb{R}^N$ with γ_N the r.v. (coordinates) x_i are independent and $\mathrm{E}x_i^2 = 1$ (as in dimension 1).

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γ_N is concentrated around the sphere $S_N(\sqrt{N})$ of radius $\sqrt{N}$.

Within this scaling factor $\sqrt{N}$, the two examples are very close to each other.

Product spaces

The concept of independent r.v.s is central in probability theory. Given probability spaces $(X_i, \mathrm{P}_i)_{i\leq N}$ there is a natural probability P on the product $X = \prod_{i\leq N} X_i$. Given sets $A_i \subset X_i$ it satisfies

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Here the large dimension effect is created by a large value of N. We will very much focus on the case where X_i is as simple as possible, $X_i = \{0, 1\}$, but for the time being we stick to the general case.

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Theorem (Mac Diarmid Inequality)

Consider a function f on X which is Lipschiz for the Hamming distance. Then for each $t > 0$ we have

$$\mathrm{P}(|f - \mathrm{E}f| \ge t) \le 2\exp(-2t^2/N).$$

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Thus the fluctuations of f around $\mathrm{E}f$ are of order $\sqrt{N}$. Hence the fluctuations of f around its median are of order $\sqrt{N}$. Consequently (as explained before) **if** $\mathrm{P}(A) \ge 1/2$ **then most points of** X **are within distance of order** $\sqrt{N}$ **of** A**.**

Scheme of proof of Mac Diarmid inequality

Lemma

Consider a function f on a probability space, $\mathrm{E}f = 0$. *Assume that* $\sup f \leq \inf f + 1$. *Then for* $t > 0$ *we have*
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Scheme of proof: Let $a = \inf f$ and $b = \sup f$. Let $\mathcal{C}$ be the convex set of functions f which satisfy the conditions $a \leq f \leq b$ and $\mathrm{E}f = 0$. On $\mathcal{C}$ the function $f \mapsto \mathrm{E} \exp tf$ is convex by Hölder's inequality.

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It attains its maximum at an extreme point. An extreme point takes only the values a and b. Then $\mathrm{E}\exp tf = \alpha \exp ta + \beta \exp tb$ with $b \leq a + 1$ and $\alpha a + \beta b = 1$. Do some calculus there. □

The proof of Mac Diarmid inequality then uses martingales difference sequences.

Continuation of proof

Define $f_N = f$ and for $n \leq N$ define recursively

$$f_n(x) = \int f_{n+1}(x) \mathrm{d}P_{n+1}(x_{n+1}).$$

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Then if f is 1-Lipschitz, $f_{n+1}(x)$ depends only on $x_1, \ldots, x_{n+1}$, and is 1-Lipschitz for the Hamming distance.

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First observation: the value of $f_{n+1}(x)$ varies by at most 1 when one changes the value of x_{n+1}.

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Second observation: When $x_1, \ldots x_n$ remain fixed the range of $f_{n+1}(x)$ is contained in an interval of length at most 1.

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Consequently this remains true for the range of $f_{n+1}(x) - f_n(x)$ because $f_n(x)$ depends only on $x_1, \ldots, x_n$.

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Consequently this remains true for the range of $f_{n+1}(x) - f_n(x)$ because $f_n(x)$ depends only on $x_1, \ldots, x_n$.

Finally, by definition, $f_n(x)$ is the average of f_{n+1} in x_{n+1} as $x_1, \ldots, x_n$ are fixed.

End of proof

Then by the lemma we have

$$\int \exp t(f_{n+1}(x) - f_n(x))\mathrm{d}P_{n+1}(x_{n+1}) \leq \exp(t^2/8).$$

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Consider the function $g(x) = \exp tf(x)$. Then using the previous inequality we obtain by induction $g_n \leq \exp((N-n)t^2/8)\exp tf_n$ and for $n = 0$

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We conclude using that for the r.v. $h = f - \mathrm{E}f$ and $u > 0$ we have $\mathrm{P}(h > u) \leq \exp(-tu)\mathrm{E}\exp th \leq \exp(Nt^2/8 - tu)$.

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Equivalently,

$$\int \exp tf_{n+1}(x)\mathrm{dP}_{n+1}(x_{n+1}) \leq \exp(t^2/8)\exp tf_n(x).$$

Consider the function $g(x) = \exp tf(x)$. Then using the previous inequality we obtain by induction $g_n \leq \exp((N-n)t^2/8)\exp tf_n$ and for $n = 0$

$$\mathrm{E}\exp tf = g_0 \leq \exp(Nt^2/2)\exp tf_0.$$

Since f_0 is constant equal to $\mathrm{E}f$ we get

$$\mathrm{E}\exp t(f - \mathrm{E}f) \leq \exp(Nt^2/8)$$

We conclude using that for the r.v. $h = f - \mathrm{E}f$ and $u > 0$ we have $\mathrm{P}(h > u) \leq \exp(-tu)\mathrm{E}\exp th \leq \exp(Nt^2/8 - tu)$.Take $t = u/N$.

A fundamental probability space

We proved that in a product space X, whenever we have a set A with $\mathrm{P}(A) \geq 1/2$ then most of the points of X are within Hamming distance of order $\sqrt{N}$ of A.

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So $I \in 2^{[N]}$ is just a subset of $\{1, \dots, N\}$.

Given $0 < p \leq 1/2$ we consider the probability μ_p on $2^{[N]}$ given by

$$\mu_p(\{I\}) = p^{\mathrm{card}\, I}(1-p)^{N-\mathrm{card}\, I}.$$

Testing the fundamental example.

This is a product measure. Denote by ν_p the probability on the two points set $\{0,1\}$ which give weight p to 1. Then the probability space $(2^{[N]},\mu_p)$ is the product space $(\{0,1\}^N,\nu_p^{\otimes N})$.

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The reason for the change of notation it that it makes easier to describe subsets of $\{0,1\}^N$.

We will make much use of the set

$$A = \{J \in 2^{[N]}; \operatorname{card} J \leq Np\}.$$

Testing, continued

The r.v. $\operatorname{card} J$ on $2^{[N]}$ is the sum of N independent r.v.s. Its expectation is Np and its variance is $Np(1-p)$, so that its distribution resembles that of $Np + \sqrt{Np(1-p)}g$ where g is standard normal.

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So there are plenty of $I \in 2^{[N]}$ with

$$\mathrm{card}\, I = Np + u\sqrt{Np(1-p)}$$

when u is not too large.

The fundamental observation

Assuming Np integer, for such a I, if $J \subset I$ and $\operatorname{card} J = Np$ then $J \in A$ and $d(I, J) = u\sqrt{Np(1-p)}$.

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But in fact we have much more: **Any** $J \subset I$ with $\operatorname{card} J \leq Np$ approaches I. Such a set J misses only $\operatorname{card} I - Np$ points of I. We can then say that each of the points i of I is, in average, missed only

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Is this a coincidence or the indication of a general phenomenon?

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- **Always study a problem in the setting with minimal structure where it makes sense.** This is the Choquet Principle.
- It often help to take a convex hull. See next slide!
- It often helps to take a product space. I spent most of my life studying product spaces.

A new way to measure the "distance" of a point to a set in a product space.

We still consider probability spaces (X_i, P_i) and the product $X = \prod_{i \leq N} X_i$ with product probability P. Give a set $A \subset X$ and $x \in X$, we define a function $\varphi(x, A)$ which in a sense measures how far x is from $A \subset X$.

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So we fix $x \in X$. For $y \in A$ define $z(x, y) \in \mathbb{R}^N$ by $z(x, y)_i = 0$ if $x_i = y_i$ and $z(x, y) = 1$ otherwise. Consider the convex hull $C(x, A)$ of the set $\{z(x, y); y \in A\} \subset \mathbb{R}^N$.

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Theorem (The Taxi Driver inequality)

We have

$$\int \exp(\frac{1}{4}\varphi(x, A)^2)\mathrm{dP}(x) \leq \frac{1}{\mathrm{P}(A)}.$$

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This no longer depends on N!

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Lemma

For each $0 \leq \lambda \leq 1$ *we have*

$$\varphi(x, A)^2 \leq (1-\lambda)^2 + (1-\lambda)\varphi(\tilde{x}, B)^2 + \lambda \varphi(\tilde{x}, A_{x_{N+1}})^2$$

continuation

Resist the desire to optimize over λ!

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Only then optimize over λ using the following:

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Consider $0 \leq r \leq 1$. *Then*

$$\inf_{0\leq\lambda\leq 1} r^{-\lambda} \exp((1-\lambda)^4/4) \leq 2 - r.$$

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Every thing falls nicely into place.

Consequences

A consequence of the Taxi Driver Inequality:

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Very useful lemma.

Lemma

Consider numbers $(\alpha_i)_{i\leq N}$, $\alpha_i\geq 0$, $\sum_{i\leq N}\alpha_i^2=1$. *Then we can find* $y\in A$ *such that*

$$\sum_{x_i\neq y_i}\alpha_i\leq\varphi(x,A).$$

Consequences of main theorem

Proof: There exists $w \in C(x, A)$ at distance $\leq \varphi(x, A)$ of the origin. By Cauchy-Schwarz we have $\sum_{i \leq N} \alpha_i w_i \leq \varphi(x, A)$. Now w is a average of $z(x, y)$ for $y \in A$. $\square$

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Taking $\alpha_i = 1/\sqrt{N}$ we obtain $d(x, A) \leq \sqrt{N}\varphi(x, A)$. Hence the inequality

$$\mathrm{P}(d(x, A) \geq t\sqrt{N}) \leq \frac{1}{\mathrm{P}(A)} \exp(-t^2/4).$$

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Recovering the situation of the fundamental example. Consider $I \in 2^{[N]}$ and take $\alpha_i = 1/\sqrt{\mathrm{card}\, I}$ for $i \in I$ and 0 otherwise. We then find $J \in A$ with $\mathrm{card}\, I \setminus J \leq \sqrt{\mathrm{card}\, J}\varphi(I, A)$. Since $\mathrm{card}\, J$ is about Np we recover the factor $\sqrt{p}$.

Convex Lipschitz functions

Theorem

Consider a convex 1-Lipschitz function f on $\mathbb{R}^N$, and P *a product measure on $[-1,1]^N \subset \mathbb{R}^N$. Then for $a \in \mathbb{R}$ and $t > 0$ we have*

$$\mathrm{P}(f \leq a)\mathrm{P}(f \geq a + t) \leq \exp(-t^2/16).$$

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This is a very precise quantitative statement that "the fluctuations of f are of order 1". In words, a convex 1-Lipschitz function behaves as well for the product measure P than a 1-Lipschitz function for the Gaussian measure.

Sketch of prooof

Let $A = \{f \leq a\}$. The key point is to show that for $x \in [-1,1]^N$ then

$$d(x, \mathrm{conv} A) \leq 2\varphi(x, A), \tag{2}$$

because then $f(x) \geq a + t \Rightarrow \varphi(x, A) \geq t/2$ and we use the previous inequality (1):

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To prove (2) note that for $x, y \in [-1,1]^N$ we have $|x_i - y_i| \leq 2z(x,y)_i$. Consider elements $y_\ell \in A$ and coefficients $\alpha_\ell \geq 0$ of sum 1. Then $\sum_\ell \alpha_\ell z(x, y_\ell) \in C(x, A)$.

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To prove (2) note that for $x, y \in [-1,1]^N$ we have $|x_i - y_i| \leq 2z(x,y)_i$. Consider elements $y_\ell \in A$ and coefficients $\alpha_\ell \geq 0$ of sum 1. Then $\sum_\ell \alpha_\ell z(x, y_\ell) \in C(x, A)$. Furthermore

$$|x_i - \sum_\ell \alpha_\ell y_{\ell,i}| \leq \sum_\ell \alpha_\ell |x_i - y_{\ell,i}| \leq 2 \sum_\ell \alpha_\ell z(x, y_\ell)_i.$$

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Reformulation: Given numbers $(t_i)_{i \leq N}$ the function $\sum_{i \leq N} t_i x_i$ on $\mathbb{R}^N$ is L-Lipschitz for $L = \sqrt{\sum_i t_i^2}$.

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Recall: In a Banach space W,

$$\|z\| = \sup\{z^*(z); z^* \in W^*, \|z^*\|^* \leq 1\}.$$

Consider vectors $(z_i)_{i\le N}$ in a Banach space. Consider the function $f\colon \mathbb{R}^N \to \mathbb{R}$ given by

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Theorem

Consider independent r.v.s U_i with $|U_i| \leq 1$. Then

$$\mathrm{P}\Big(\big|\|\sum_{i\leq N} U_i x_i\| - M\big| \geq t\Big) \leq 2\exp(-t^2/(16\sigma^2)) \ .$$

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The most important case is $U_i = \varepsilon_i$, independent random signs.

Control by several points

Given a point x in a product space X and a set $A \subset X$ we have tried to find a point $y \in A$ such that many of the coordinates y_i and x_i coincide. Now, we are given sets $A_1, \dots, A_q$ and we try to find points $y_\ell \in A_\ell$ such for many coordinates $x_i \in \{y_{1,i}, \dots, y_{q,i}\}$.

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We define $\varphi(x, A_1, \dots, A_q)$ as the minimum over $y_1 \in A_1, \dots, y_q \in A_q$ of $\operatorname{card}\{i \le N; x_i \notin \{y_{1,i}, \dots y_{q,i}\}\}$.

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Theorem

we have

$$\int q^{\varphi(x,A_1,\dots A_q)} \mathrm{dP}(x) \le \frac{1}{\prod_{\ell \le q} \mathrm{P}(A_\ell).}$$

This is again proved by induction over N so that it can't be hard!

Discussion

In particular if $\mathrm{P}(A_\ell) \geq 1/2$ for each ℓ, then we have

$$\mathrm{P}(\varphi(x, A_1, \ldots, A_q) \geq k) \leq \frac{2^q}{q^k}$$

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$$\mathrm{P}(\varphi(x, A_1, \ldots, A_q) \geq n) \geq p^N = N^{-N}$$

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whereas the upper bound given by the theorem is $e^q q^{-N} = \exp(q - N \log q)$. This is of the right order for q a small proportion of $N/\log N$.

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Application to Probability in Banach spaces, II

We write $\|\sum_i \varepsilon_i Z_i\| = I + II$ where

$$I = \|\sum_i \varepsilon_i Z_i\| - \mathrm{E}_\varepsilon \|\sum_i \varepsilon_i Z_i\|$$

$$II = \mathrm{E}_\varepsilon \|\sum_i \varepsilon_i Z_i\|.$$

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One is reduced to the study of the r.v.s $\mathrm{E}_\varepsilon \|\sum_i \varepsilon_i Z_i\|$.

More details

Let P_i be the law of Z_i. We consider the probability space $(\Omega, P) = \prod_{i\in I}(W, P_i)$. A generic point is denoted by $(x_i)_{i\in I}$.

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$$\psi(x) = E_\varepsilon \| \sum_{i\in I} \varepsilon_i x_i \|.$$

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Lemma

We have

$$P(\psi(x) \geq qC + k(x)) \leq \frac{2^q}{q^k}.$$

Is this lecture ever going to end?

It suffices to prove that

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disjoint sets $J_1, \dots, J_q$ such that $i \in J_\ell \Rightarrow x_i = y_{j_\ell, i}$ and
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$$\mathrm{E}\left\|\sum_{i\in I} \varepsilon_i x_i\right\| \leq \sum_{\ell\leq q} \mathrm{E}\left\|\sum_{i\in J_\ell} \varepsilon_i x_i\right\| + \sum_{i\in I\setminus\cup J_\ell} \|x_i\|.$$

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This is less that $qC + k(x)$.